



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

- I ✓ Explain spontaneous and non-spontaneous processes.
- II ✓ Describe the most probable velocity.
- III ✓ How can you determine the pre-exponential factor from the Arrhenius equation/plot?
- IV ✓ What is the difference between isolated and closed systems?
- V ✓ Define open system.
- VI ✓ Give two statements of the 2nd law of thermodynamics.
- VII ✓ What is the "Partition Function"?
- VIII ✓ How do ionic reactions differ from covalent reactions?
- IX ✓ Define the term microstate.
- X ✓ Write two different forms of the Arrhenius equation.
- XI ✓ What is adiabatic demagnetization?
- XII ✓ How does the barometric formula describe the variation of atmospheric pressure with altitude?
- XIII ✓ What are fast reactions?
- XIV ✓ What factors influence the rate of a bimolecular reaction?
- XV ✓ How does the concept of entropy relate to the second law of thermodynamics?

Answer the following questions.

(3x10=30)

- Q.2a ✓ Write a note on Maxwell's law of distribution of velocities. 15-20
- ✓ Prove that $Q = Q_t \cdot Q_r \cdot Q_v \cdot Q_e$ 272
- Q.3a ✓ State and explain different statements of the 3rd law of thermodynamics. 236
- ✓ Drive statistically expression for equilibrium constant using Partition Function. 286
- Q.4a ✓ Write a note on transition state theory. 443
- ✓ What is the effect of temperature on the rate of reaction? 434

what is difference b/w Spontaneous and non Spontaneous process ?

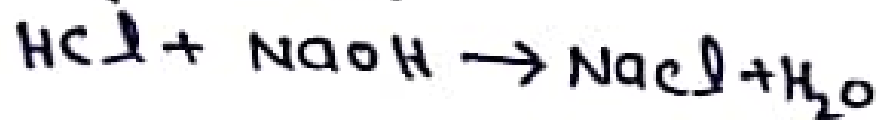
Spontaneous Process

A process which takes place on its own without any outside assistance and move towards an equilibrium state is called Spontaneous process.

Example:-

Flow of heat from hot body to cold body.

Reaction b/w Strong acid and Strong base.



Non Spontaneous Process

A process which is the reverse of spontaneous and does not take place on its own is called non spontaneous process.

Example:-

Freezing of water in refrigerator.



Describe Most Probable velocity ?

The most probable velocity is defined as the velocity possessed by maximum number of molecules of gas at given temperature.

The expression for most probable velocity can be derived mathematically by using condition of maxima by setting first derivatives of $\frac{1}{n} \cdot \frac{dn_c}{dc}$ with respect to speed equal to zero.

The expression for most probable velocity is

$$C_{mp} = 1.414 \sqrt{\frac{RT}{M}}$$

How can you determine pre-exponential factor from Arrhenius equation?

The Arrhenius equation is

$$K = Ae^{-\frac{E_a}{RT}}$$

where,

"K" is the rate constant,

"A" is pre-exponential factor

" E_a " is the Energy of Activation

"R" is universal gas constant

"T" is Temp. in Kelvin,

Take natural log we can solve for "A"

$$\ln K = \ln A e^{-\frac{E_a}{RT}}$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

$$\ln K = -\frac{E_a}{RT} + \ln A$$

Comparing eq—

$$Y = MX + C$$

$$y = \ln K, \quad M = -\frac{E_a}{R}, \quad X = \frac{1}{T}, \quad C = \ln A$$

So this way plotting $\ln K$ versus $\frac{1}{T}$ makes it experimentally possible to determine "A"

Another way is

$A = \frac{K}{e^{-\frac{E_a}{RT}}}$

what are closed and isolated system?

closed system

if a system can exchange only energy with the surroundings but no exchange of matter are called closed system.

Example :-

Water placed in closed metallic vessel.

isolated system

if a system can neither exchange energy nor matter with the surroundings are called isolated system.

Example :-

Thermos flask.

Open system: If a system can exchange both matter and energy with surroundings. For example, animals and plants.

Give Two Statements of 2nd law of thermodynamics?

Kelvin-Planck Statement:-

No process is possible in which the sole result is the absorption of heat from reservoir and its complete conversion into work.

OR

No heat engine can convert all of the heat it receives, into useful work without producing some waste heat.

Clausius Statement:-

Heat can not spontaneously flow from colder body to hotter body without external work being performed on the system.

This statement emphasizes the direction of heat transfer and highlights that natural processes tend to increase the overall entropy of a closed system.

Define the Term Partition Function?

The sum of the energy of all the states of the system is called the Partition Function. It can also be stated as the way in which energy is partitioned among the molecules constituting the system is called Partition function.

$$Q = \sum_{i=0}^{\infty} g_i e^{-\frac{\epsilon_i}{kT}}$$

Ionic vs. Covalent Reactions

Ionic Reactions

- Involve transfer of electrons.
- Occur between metals and non-metals.
- Often fast and occur in aqueous solutions.

Covalent Reactions

- Involve sharing of electrons.
- Occur between non-metals.
- Can occur in various phases.

Define the Term Micro State ?

A microstate is defined as "A state of the system where the distribution of particles over the energy levels is specified."

A microstate refers to the specific arrangement of individual particles within a system. are called microstate. it characterizes the exact state of system by describing the positions and momenta of all its constituent particles.

Example :

An example of microstate could be the exact position and velocities of all gas molecules in a container at given moment.

rite Two different forms of Arrhenius equations?

We know that the Arrhenius equations

$$K = Ae^{-E_a/RT}$$

Take $\ln$ on B/s

$$\ln K = \ln(Ae^{-E_a/RT})$$

$$\ln K = \ln A + \ln e^{-E_a/RT}$$

$$\ln K = \ln A - \frac{E_a}{RT} \ln e$$

$$(\because \ln e = 1)$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

This eq. is linear form of Arrhenius equation.

Let the Temp. T_1 and rate constant for a reaction is K_1 the above eq. will be

$$\ln K_1 = \ln A - \frac{E_a}{RT_1} \rightarrow \textcircled{A}$$

Similarly at Temp. T_2 and rate constant will be K_2 then eq. will be

$$\ln K_2 = \ln A - \frac{E_a}{RT_2} \rightarrow \textcircled{B}$$

Subtract eq. $\textcircled{A}$ and $\textcircled{B}$

$$\ln K_2 - \ln K_1 = \cancel{\ln A} - \frac{E_a}{RT_2} - \cancel{\ln A} + \frac{E_a}{RT_1}$$

$$\ln \frac{K_2}{K_1} = -\frac{E_a}{RT_2} + \frac{E_a}{RT_1}$$

$$\ln \frac{K_2}{K_1} = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

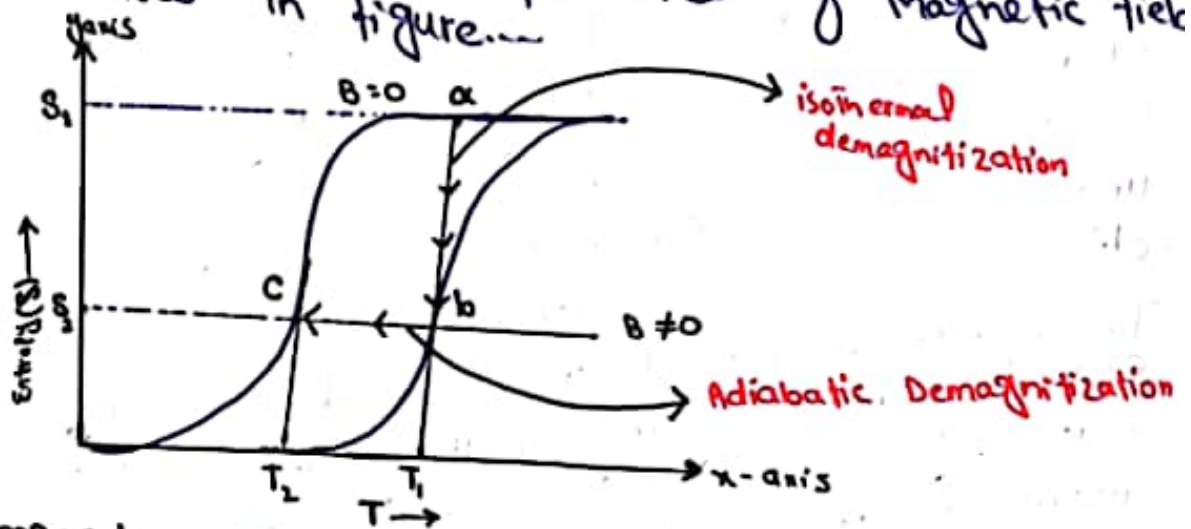
$$\ln \frac{K_2}{K_1} = \frac{E_a}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

this is another form of Arrhenius equation

What is adiabatic Demagnetization? Give Example?

According to 3rd law of thermodynamics the entropy of system decrease with decrease in temperature and ultimately become zero at absolute zero for perfect crystalline solid.

Paramagnetic substance w.r.t temp. in the absence and presence of magnetic field "B" are shown in figure.



At high Temperature, Entropy of system is maximum and independent of magnetic field. But in low Temp-Range entropy depends upon magnetic field.

Adiabatic Demagnetization is a process/technique which can be used to get the system at very low Temp. about 0.05K Temp. can be obtained by using this way.

We can see above diagram in the path "b-c" show the adiabatic demagnetization during which entropy of system remains constant but Temp. of system decreases from "T1 to T2", this process is called is-entropic process.

When we apply external M.F. on paramagnetic substance surrounded by helium gas at constant Temp. then its entropy decreases along path "a-b". The path a-b is an isothermal process during which entropy of paramagnetic substance decreases but Temp. of system remains constant. This process is called isothermal demagnetization.

Effect of Altitude (Height) on the Vertical distribution of particles

According to equation (1.53), the pressure exerted by a gas column decreases exponentially with the altitude. Thus a plot of $\ln(P/P_0)$ against height (h) will be a straight line with slope equal to $-Mg/R$ provided T remains constant. One of the characteristics of an exponential decay is that the relative decrease in pressure $-dP/P$ is directly proportional to dh . Thus, if pressure decreases by a factor of 2 in going up by a distance h , it will further decrease by the same factor for another height of h .

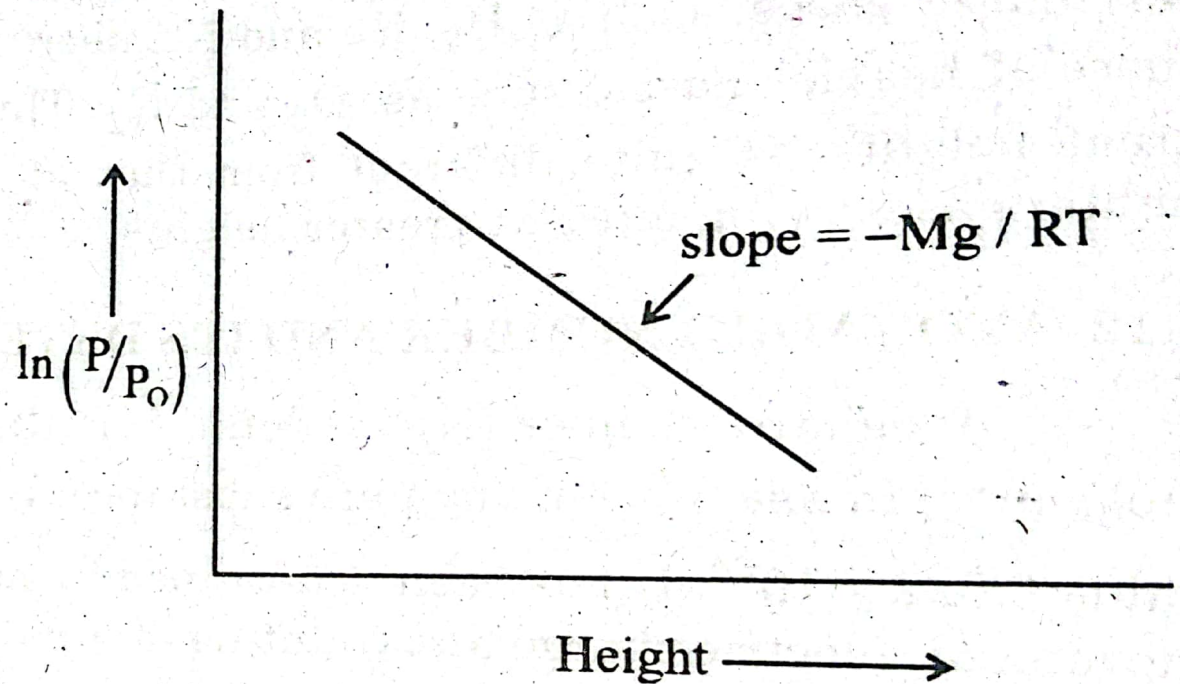


Fig. 1.9. Plot of $\ln(P/P_0)$ versus height (h)

What are Fast Reactions?

Fast Reactions are chemical Reaction that occur at very fast Rate, in these Reaction Reactant Quickly transform into product often within fraction of Second or even faster. Fast Reaction as name indicates are very high speed Reactions and exhibits very high value of Rate constant and very short half life periods.

Such Reactions may exemplified by solution Phase Reaction involving ionic species, such as



Some advanced techniques such as "flow method" "Pulse method" are introduced to study the kinetics of such Reactions. and Types of Fast Reactions are "chemical, Biological, Physical Nuclear Reactions."

Factors Influencing Bimolecular Reaction Rates

Primary Factors

- **Concentration:** Higher concentration leads to more frequent collisions.
- **Temperature:** Increased temperature increases kinetic energy, leading to more collisions with sufficient energy.
- **Orientation:** Molecules must collide with correct orientation for reaction.
- **Nature of reactants:** Bond strengths and molecular complexity affect reaction rate.
- **Presence of a catalyst:** Lowers activation energy, increasing reaction rate.

Additional Factors (Especially for Ionic Reactions)

- **Solvent effect:** Nature of solvent (polarity

Additional Factors (Especially for Ionic Reactions)

- **Solvent effect:** Nature of solvent (polarity, viscosity) affects reaction rate.
- **Hydrostatic pressure:** Affects reactions involving gases or liquids.
- **Dielectric permittivity of solvent:** Influences ionic reactions by affecting ion-ion interactions.
- **Ionic strength:** Affects rates of ionic reactions due to shielding effects.

Entropy and the Second Law of Thermodynamics

Entropy is a measure of the disorder or randomness of a system. The higher the entropy, the more disordered the system.

The Second Law of Thermodynamics states that the total entropy of an isolated system always increases over time or, at best, remains constant. This means that natural processes tend to move towards a state of greater disorder.

Entropy as a measure of disorder: When a system undergoes a change, it tends to move towards a state of higher entropy, meaning more disorder. This is a natural tendency.

The Second Law as a consequence: The Second Law is essentially a statement about this tendency towards increased disorder. It quantifies this observation by saying that the total entropy of a system can never decrease.